Isolation Bearing Design

202 Rate Independent Damping

Example 04 - rivtbook

proj. 0004

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3.10 HARMONIC VIBRATION WITH RATE-INDEPENDENT DAMPING [1]

3.10.1 Rate-Independent Damping

Experiments on structural metals indicate that the energy dissipated internally in cyclic straining of the material is essentially independent of the cyclic frequency. Similarly, forced vibration tests on structures indicate that the equivalent viscous damping ratio is roughly the same for all natural modes and frequencies. Thus we refer to this type of damping as rate-independent linear damping. Other terms used for this mechanism of internal damping are structural damping, solid damping, and hysteretic damping. We prefer not to use these terms because the first two are not especially meaningful and the third is ambiguous because hysteresis is a characteristic of all materials or structural systems that dissipate energy.

Rate-independent damping is associated with static hysteresis due to plastic strain, localized plastic deformation, crystal plasticity, and plastic flow in a range of stresses within the apparent elastic limit. On the microscopic scale the inhomogeneity of stress distribution within crystals and stress concentration at crystal boundary intersections produce local stress high enough to cause local plastic strain even though the average “global” stress may be well below the elastic limit. This damping mechanism does not include the energy dissipation in global plastic deformations, which, as mentioned earlier, is handled by a nonlinear relationship between force fS and deformation u.

The simplest device that can be used to represent rate-independent linear damping is to assume that the damping force is proportional to velocity and inversely proportional to frequency:

$fD = (\eta k/\omega)\ \dot{u}$ … (3.10.1)

where k is the stiffness of the structure and η is a damping coefficient. The energy dissipated by this type of damping in a cycle of vibration at frequency ω is independent of ω (Fig. 3.10.1). It is given by Eq. (3.8.1) with c replaced by η*k/ω:

$ED = \pi\eta k u02 = 2\pi\eta{*}E\_So$ … (3.10.2)

In contrast, the energy dissipated in viscous damping increases linearly with the forcing frequency, as shown in Fig. 3.10.1.

Rate-independent damping is easily described if the excitation is harmonic and we are interested only in the steady-state response of this system. Difficulties arise in translating this damping mechanism back to the time domain. Thus it is most useful in the frequency-domain method of analysis (Section 1.10.3). The complex term k(1 + iη)*u represents both the elastic and damping forces at the same time; k(1 + iη) is the complex stiffness of the system. There are advantages in this terminology if the reader is familiar with complex-variable notation. The complex stiffness has no physical meaning, however, in the same engineering sense as the elastic stiffness.

[Figure 3.10.1 - Energy dissipated in viscous damping and rate-independent damping. Plot of energy dissipated ED (vertical axis) versus forcing frequency ω (horizontal axis). A straight line through the origin, labeled “Viscous damping,” rises linearly with ω. A horizontal line, labeled “Rate-independent damping,” stays constant with ω. The two lines cross at ω = ωn. Image file: images/fig_3.10.1_energy_dissipated.png]

3.10.2 Steady-State Response to Harmonic Force

The equation for an SDF system with rate-independent linear damping, denoted by a crossed box in Fig. 3.10.2, is Eq. (3.2.1) with the damping term replaced by Eq. (3.10.1):

$m\ddot{u} + (\eta k/\omega)\dot{u} + ku = p(t)$ … (3.10.3)

The mathematical solution of this equation is quite complex for arbitrary p(t). Here we consider only the steady-state motion due to a sinusoidal forcing function, p(t) = p0sin(ωt), which is described by

u(t) = u0 sin(ωt - φ) … (3.10.4)

The amplitude u0 and phase angle φ are

u0 = (u_st)0 1 / sqrt((1 - (ω/ωn)2)2 + η2) … (3.10.5)

φ = arctan(η / (1 - (ω/ωn)2)) … (3.10.6)

These results are obtained by modifying the viscous damping ratio in Eqs. (3.2.11) and (3.2.12) to reflect the damping force associated with rate-independent damping, Eq. (3.10.1). In particular, ζ was replaced by

ζ = c/c_c = (ηk/ω) / (2mωn) = η / (2(ω/ωn)) … (3.10.7)

Shown in Fig. 3.10.3 by solid lines are plots of u0/(u_st)0 and φ as a function of the frequency ratio ω/ωn for damping coefficient η = 0, 0.2, and 0.4; the dashed lines are described in the next section. Comparing these results with those in Fig. 3.2.6 for viscous damping, two differences are apparent: First, resonance (maximum amplitude) occurs at ω = ωn, not at ω < ωn. Second, the phase angle for ω = 0 is φ = arctan(η) instead of zero for viscous damping; this implies that motion with rate-independent damping can never be in phase with the forcing function.

These differences between forced vibration with rate-independent damping and forced vibration with viscous damping are not significant, but they are the source of some difficulty in reconciling physical data. In most damped vibration, damping is not viscous, and to assume that it is without knowing its real physical characteristics is an assumption of some error. In the next section this error is shown to be small when the real damping is rate independent.

[Figure 3.10.2 - SDF system with rate-independent linear damping. Schematic: a mass m rests on a friction-free surface (shown with rollers), connected to a fixed wall by a spring of stiffness k and, in parallel, a crossed-box symbol labeled η (representing the rate-independent damping element). The mass displaces by u (arrow pointing right, labeled “u”) under applied force p(t) (arrow pointing right, labeled “p(t)”). Image file: images/fig_3.10.2_sdf_system.png]

[Figure 3.10.3 - Response of system with rate-independent damping: exact solution and approximate solution using equivalent viscous damping. Top plot: Deformation response factor Rd = u0/(u_st)0 (vertical axis, 0 to 5) versus frequency ratio ω/ωn (horizontal axis, 0 to 3). Solid curves show rate-independent damping for η = ζ = 0 (sharp peak at Rd = 5 at ω/ωn = 1), ζ = 0.1 with η = 0.2 (lower peak), and ζ = 0.2 with η = 0.4 (peak near Rd = 2.5). Dashed curves show the equivalent-viscous-damping approximation overlaid on each case. Bottom plot: Phase angle φ (vertical axis, 0° to 180°) versus the same frequency ratio ω/ωn. Curves rise from a small angle at ω/ωn = 0, pass through 90° at ω/ωn = 1, and approach 180° for large ω/ωn, again with solid (rate-independent) and dashed (equivalent viscous) curves for $\eta = \zeta = 0$, $\eta = 0.2/\zeta = 0.1$, and $\eta = 0.4/\zeta = 0.2$. Image file: images/fig_3.10.3_response_factor.png]

3.10.3 Solution Using Equivalent Viscous Damping

In this section an approximate solution for the steady-state harmonic response of a system with rate-independent damping is obtained by modeling this damping mechanism as equivalent viscous damping.

Matching dissipated energies at $\omega = \omega n$ (Fig. 3.10.1) led to Eq. (3.9.2), where ED is given by Eq. (3.10.2), leading to the equivalent viscous damping ratio:

$\zeta\_eq = \eta/2$ … (3.10.8)

Substituting this ζ_eq for ζ in Eqs. (3.2.10) to (3.2.12) gives the system response. The resulting amplitude u0 and phase angle φ are shown by the dashed lines in Fig. 3.10.3. This approximate solution matches the exact result at $\omega = \omega n$ because that was the criterion used in selecting ζ_eq. Over a wide range of excitation frequencies the approximate solution is seen to be accurate enough for many engineering applications. Thus Eq. (3.10.3) - which is difficult to solve for arbitrary force p(t) that contains many harmonic components of different frequencies ω - can be replaced by the simpler Eq. (3.2.1) for a system with equivalent viscous damping defined by Eq. (3.10.8). This is the basic advantage of equivalent viscous damping.

3.11 HARMONIC VIBRATION WITH COULOMB FRICTION

3.11.1 Equation of Motion

Shown in Fig. 3.11.1 is a mass-spring system with Coulomb friction force $F = \mu * N$ that opposes sliding of the mass. As defined in Section 2.4, the coefficients of static and kinetic friction are assumed to be equal to μ, and N is the normal force across the sliding surfaces. The equation of motion is obtained by including the exciting force in Eqs. (2.4.1) and (2.4.2) governing the free vibration of the system:

$m\ddot{u} + ku \pm F = p(t)$ … (3.11.1)

The sign of the friction force changes with the direction of motion; the positive sign applies if the motion is from left to right ($\dot{u} > 0$) and the negative sign is for motion from right to left ($\dot{u} < 0$). Each of the two differential equations is linear, but the overall problem is nonlinear because the governing equation changes every half-cycle of motion. Therefore, exact analytical solutions would not be possible except in special cases.

[Figure 3.11.1 - SDF system with Coulomb friction. Schematic: a mass m, connected to a fixed wall by a spring of stiffness k, rests on a surface exerting friction force ±F. The mass is subjected to applied force p(t) (arrow pointing right, labeled “p(t)”). Image file: images/fig_3.11.1_coulomb_friction_sdf.png]

3.11.2 Steady-State Response to Harmonic Force

An exact analytical solution for the steady-state response of the system of Fig. 3.11.1 subjected to harmonic force was developed by J. P. Den Hartog in 1933. The analysis is not included here, but his results are shown by solid lines in Fig. 3.11.2; the dashed lines are described in the next section. The displacement amplitude u0, normalized relative to (u_st)0 = p0/k, and the phase angle φ are plotted as a function of the frequency ratio ω/ωn for three values of F/p0. If there is no friction, F = 0 and u0/(u_st)0 = (Rd)_ζ=0, the same as in Eq. (3.1.11) for an undamped system. The friction force reduces the displacement amplitude u0, with the reduction depending on the frequency ratio ω/ωn.

[Figure 3.11.2 - Deformation response factor and phase angle of a system with Coulomb friction excited by harmonic force. Exact solution from J. P. Den Hartog; approximate solution is based on equivalent viscous damping. Top plot: Deformation response factor Rd = u0/(u_st)0 (vertical axis, 0 to 3.5) versus frequency ratio ω/ωn (horizontal axis, 0 to 2.0). Solid curves (Coulomb friction) are shown for F/p0 = 0, 0.5, 0.7, π/4, and 0.8, each rising to a peak near ω/ωn = 1 and diverging to infinity at ω/ωn = 1 for the lower friction values; the F/p0 = 0.8 curve shows a rounded peak around Rd ≈ 1.6 near ω/ωn ≈ 0.85 and then drops off. Dashed curves (equivalent viscous damping) approximate each solid curve. Bottom plot: Phase angle φ (vertical axis, 0° to 180°) versus the same frequency ratio. Curves for F/p0 = 0 (φ = 0 for ω/ωn < 1, jumping to ~180° above resonance), and for F/p0 = 0.5, 0.7, 0.8, with corresponding dashed equivalent-viscous-damping approximations. Image file: images/fig_3.11.2_coulomb_response.png]

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[1] Anil K.Anil K. Chopra, Dynamics of Structures: Theory and Applications toEarthquake Engineering. Englewood Cliffs, NJ, USA: Prentice Hall, 1995.